

Sarah Connor

AI Security Engineer · Adversarial ML & Red Teaming
sarah@sconnor.dev · Austin, TX (US) · +1 123 456 7890
github.com/sconnor · linkedin.com/in/sarahconnor · sconnor.dev

SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

EXPERIENCE

Senior AI Security Engineer
Cyberdyne Systems

2023 – Present
Austin, USA

Leading red-team engagements against internal LLM-powered products before launch.

- Built an automated prompt-injection fuzzing harness that surfaced 40+ high-severity jailbreaks across 6 production LLM features prior to release.
- Led adversarial robustness reviews for a fraud-detection model serving 2M inference requests/day, reducing evasion success rate from 18% to under 2%.
- Designed and shipped a model-extraction detection pipeline (query-pattern anomaly scoring) now running across all customer-facing inference endpoints.

Python · PyTorch · Burp Suite · Kubernetes · OPA/Gatekeeper

Security Engineer, Applied ML
Skynet Analytics

2021 – 2023
Remote

Owned the security review process for the ML platform team’s model deployment pipeline.

- Authored the org’s first threat model for LLM-integrated services, adopted as the standard review template company-wide.
- Cut mean time to patch critical model-serving CVEs from 21 days to 4 by embedding security gating directly into the MLOps CI/CD pipeline.
- Ran quarterly red-team exercises against internal chatbots, discovering data-exfiltration paths later closed via output filtering and scoped tool permissions.

Terraform · AWS · MLflow · Semgrep · Snyk

Machine Learning Engineer
Tyrell ML

2019 – 2021
Ljubljana, Slovenia

Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.

- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company’s core payments pipeline.
- Designed the team’s first automated retraining pipeline, cutting model refresh time from weeks to days.

TensorFlow · scikit-learn · Docker

SKILLS

Security	Threat modeling, Burp Suite, Nmap, Metasploit
ML/AI	PyTorch, vLLM, adversarial robustness, prompt-injection defense
Infra	Kubernetes, Terraform, AWS, CI/CD pipeline hardening
Practice	Incident response, vulnerability research, red-team exercise design

>	EDUCATION	
	M.S. Computer Security	2023
	Stanford University	Stanford, USA
	B.S. Computer Science	2019
	University of Ljubljana	Ljubljana, Slovenia
>	PROJECTS	
	promptfirewall	2023
	github.com/sconnor/promptfirewall	
>	CERTIFICATIONS, AWARDS & PUBLICATIONS	
	OSCP – Offensive Security Certified Professional	2019
	Offensive Security	
	AI Village CTF – 1st Place	2024
	DEF CON	
	Detecting Prompt Injection at Scale	2024
	arxiv.org/abs/2403.09217	